ASSIGNMENT 1

1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Ans:- • **Frontend Development:** Refers to the client side of web development. It focuses on everything the user sees, touches, and interacts with directly in their browser (UI/UX layout, buttons, forms, animations).

- *Tools/Languages:* HTML, CSS, JavaScript, React, Vue, Angular.
- *Real-World Example:* The user interface of **Netflix**, where you scroll through movie posters, play trailers, and click buttons.

• **Backend Development:** Refers to the server side of web development. It handles core application logic, database management, user authentication, security, and server configuration.

- *Tools/Languages:* Node.js, Python (Django/Flask), Java, PostgreSQL, MongoDB.
- *Real-World Example:* The processing backend of **Netflix** that verifies your subscription, fetches your watch history from the database, and streams the correct video file to your browser.

• **Full-Stack Development:** Involves working on both the frontend and backend layers of an application. A full-stack developer is capable of building an entire web application end-to-end, from user interface design to database schema management.

- *Real-World Example:* Building an entire e-commerce store (like **Amazon**) — designing the product listing page (Frontend) as well as writing the payment processing and inventory update API endpoints (Backend).

2. Create a simple diagram showing how the client-server model works in web architecture.

Ans:- [Client / Browser] --- (1) HTTP Request ---> [Web Server]

<-- (2) HTTP Response ---

3. Describe how a browser requests and displays a web page from a web server.

Ans:- • **DNS Lookup:** Resolves the URL domain name to an IP address.

• **HTTP Request:** Sends a request to the target server for page files.

- **Server Response:** The server sends back HTML, CSS, JavaScript, and assets.
- **Rendering:** The browser parses HTML to DOM, applies CSS styles, and executes JavaScript to display the final page.

4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

Ans:- • **Code Editor (VS Code):** Writing and debugging source code.

- **Web Browser (Chrome/Firefox):** Rendering web pages and inspecting DOM/network requests.
- **Version Control (Git):** Tracking code history and collaborating.
- **Runtime Environment (Node.js):** Executing JavaScript on the local machine outside the browser.

5. Explain what a web server is and give examples of commonly used servers.

Ans:- A **web server** is software/hardware that stores web assets and serves them to clients over HTTP/HTTPS upon request.

- **Examples:** Apache HTTP Server, Nginx, Microsoft IIS, Express (Node.js).

6. Define the roles of a frontend developer, backend developer, and database administrator in a project.

Ans:- • **Frontend Developer:** Builds the visual layout, interactive UI elements, and user experience.

- **Backend Developer:** Constructs server architecture, database integrations, and API endpoints.
- **Database Administrator (DBA):** Manages, secures, optimizes, and maintains database systems and data structures.

7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.


```html
<!DOCTYPE html>
<html lang="en">
<head>
    <!-- <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0"> -->
    <title>Document</title>
</head>
<body>
    <h1>Heading</h1>
    <p><b>Hello</b> This is nothing<i> Italic</i></p>
    <hr>
    <strong>Vhaghar</strong>
    <em>Italic</em>
    <h1>H<sub>2</sub>O</h1>

    <h1>(a+b)<sup>2</sup> = a<sup>2</sup> + 2ab + b<sup>2</sup></h1>
    <h1>price Rs : <del>100</del> <ins>80</ins></h1>
    <ul type="square">
        <li>HTML</li>
        <li>CSS</li>
        <li>JS</li>
    </ul>
    <ol type = "1" start = "5">
        <li>HTML</li>
        <li>CSS</li>
        <li>JS</li>
    </ol>
```

8. Explain the difference between static and dynamic websites. Provide an example of each.

Ans:- • **Static:** Fixed content served as-is to every user (e.g., a simple portfolio website).

• **Dynamic:** Content updates in real time based on user interactions, database data, or user accounts (e.g., Twitter feed).

9. Research and list five web browsers. Explain how rendering engines differ between them.

Ans:- • **Google Chrome:** Blink engine.

• **Apple Safari:** WebKit engine.

• **Mozilla Firefox:** Gecko engine.

• **Microsoft Edge:** Blink engine.

• **Brave:** Blink engine.

• **Difference:** Engines interpret HTML/CSS and execute JS differently, leading to subtle variations in rendering speed, CSS support, and memory optimization across browsers.

10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

Ans:- [Client / Browser]

|

HTTP | API Request

▼

[API / Web Server]

|

Database | Queries (SQL/NoSQL)

▼

[(Database)]